

Week 1: Requirement Analysis and Customer Database Module

1. Requirement Analysis

Understanding the E-Commerce System

An e-commerce platform allows customers to search for products, select the required items, place orders, make payments, and receive their purchases. A database is required to maintain important information about customers, products, sellers, orders, payments, and reviews. Properly organized data helps the system manage customer activities and transactions efficiently.

The Process Flow

- **Product Search:** The customer explores the available products and searches for the required item.
- **Cart Management:** The selected products are added to the shopping cart before placing the order.
- **Order Confirmation:** The customer provides the necessary delivery details and confirms the order.
- **Payment:** The customer selects an available payment method and completes the payment process.
- **Product Delivery:** Once the order is processed, the product is prepared for dispatch and delivered to the customer's address.
- **Feedback and Assistance:** After receiving the product, the customer can provide feedback or contact support regarding returns, replacements, or other issues.

Entities:

Entity	Description	Key Attributes	Key Relationships
Customer	Stores information about people who use the platform to purchase products.	Customer _id (PK), first _name, last _name, email, phone, created _at	One-to-Many with Order and Review
Seller	Represents the vendor responsible for offering products through the platform.	Seller _id (PK), company _name, email, phone, rating, bank _details	One-to-Many with Product
Product	Contains details of the products available for customers to purchase.	Product _id (PK), seller _id (FK), title, description, price, stock _quantity, category	Many-to-One with Seller, One-to-Many with Review
Order	Records the purchase made by a customer through the platform.	Order _id (PK), customer _id (FK), order _date, status, total _amount, shipping _address	Many-to-One with Customer, One-to-One with Payment

Payment	Maintains the payment information related to a customer's order.	Payment_id (PK), order_id (FK), payment_method, status, transaction_id, amount, payment_date	One-to-One with Order
Review	Stores the rating and comments provided by customers about purchased products.	Review_id (PK), customer_id (FK), product_id (FK), rating, comment, created_at	Many-to-One with Customer and Product

Customer Table Creation

The Customer table is created to maintain the basic details of users registered on the e-commerce platform. A unique customer ID is used to identify each customer record.

```
SQL> create table customer(customer_Id int primary key, customer_Name varchar(30) not null, Email varchar(30), Phone varchar(30) unique, Address varchar(30), Registration_Date Date);
Table created.
```

Describe Customer Table

After creating the Customer table, the DESCRIBE command is used to check the structure of the table. It displays the column names, data types, and other details associated with each attribute.

```
SQL> desc customer;
Name                               Null?    Type
-----
CUSTOMER_ID                       NOT NULL NUMBER(38)
CUSTOMER_NAME                     NOT NULL VARCHAR2(30)
EMAIL                             VARCHAR2(30)
PHONE                             VARCHAR2(30)
ADDRESS                           VARCHAR2(30)
REGISTRATION_DATE                 DATE
```

Insert Values

The INSERT command is used to add customer information into the Customer table. Multiple records can be entered based on the number of customers required for the database.

```
SQL> insert into customer values(001,'Dhoni','Dhoni77@gmail.com','9876543210','Chennai',date'2021-04-21');
1 row created.
```

```
SQL> INSERT ALL
2 INTO customer VALUES (002, 'Virat', 'Virat18@gmail.com', '9876543123', 'Bangalore', DATE '2022-06-24')
3 INTO customer VALUES (003, 'Ashwin', 'ashwin99@gmail.com', '9876543212', 'Chennai', DATE '2021-09-11')
4 INTO customer VALUES (004, 'SanjuSamson', 'sanju11@gmail.com', '9876543234', 'Kerala', DATE '2021-09-11')
5 INTO customer VALUES (005, 'Shreyas Iyer', 'shreyas96@gmail.com', '9876545689', 'Punjab', DATE '2025-03-27')
6 INTO customer VALUES (006, 'Hardhik Pandya', 'hardhik33@gmail.com', '9876598765', 'Hyderabad', DATE '2019-03-30')
7 INTO customer VALUES (007, 'Ruturaj Gaikwad', 'ruturaj31@gmail.com', '983214561', 'Chennai', DATE '2020-03-20')
8 INTO customer VALUES (008, 'Rohit Sharma', 'rohit45@gmail.com', '9876512340', 'Bangalore', DATE '2018-09-14')
9 INTO customer VALUES (009, 'Raina', 'raina3@gmail.com', '9812345390', 'Chennai', DATE '2019-11-12')
10 INTO customer VALUES (010, 'Jadeja', 'jadeja8@gmail.com', '9123456781', 'Chennai', DATE '2021-07-12')
11 SELECT * FROM dual;
9 rows created.
```

Display Customer Records

The stored customer information can be viewed using the SELECT command.

```
SQL> select*from customer;
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
1 Dhoni            Dhoni77@gmail.com
9876543210      Chennai            21-APR-21
2 Virat            Virat18@gmail.com
9876543123      Bangalore          24-JUN-22
3 Ashwin           ashwin99@gmail.com
9876543212      Chennai            11-SEP-21
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
4 SanjuSamson      sanju11@gmail.com
9876543234      Kerala             11-SEP-21
5 Shreyas Iyer     shreyas96@gmail.com
9876545689      Punjab             27-MAR-25
6 Hardhik Pandya   hardhik33@gmail.com
9876598765      Hyderabad          30-MAR-19
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
7 Ruturaj Gaikwad  ruturaj31@gmail.com
983214561      Chennai            20-MAR-20
8 Rohit Sharma     rohit45@gmail.com
9876512340      Bangalore          14-SEP-18
9876512340
10 rows selected.
```

```
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
4 SanjuSamson      sanju11@gmail.com
9876543234      Kerala             11-SEP-21
5 Shreyas Iyer     shreyas96@gmail.com
9876545689      Punjab             27-MAR-25
6 Hardhik Pandya   hardhik33@gmail.com
9876598765      Hyderabad          30-MAR-19
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
7 Ruturaj Gaikwad  ruturaj31@gmail.com
983214561      Chennai            20-MAR-20
8 Rohit Sharma     rohit45@gmail.com
9876512340      Bangalore          14-SEP-18
9 Raina            raina3@gmail.com
9812345390      Chennai            12-NOV-19
CUSTOMER_ID  CUSTOMER_NAME      EMAIL
-----
PHONE-----
ADDRESS-----
REGISTRATION_DATE-----
10 Jadeja          jadeja8@gmail.com
9123456781      Chennai            12-JUL-21
10 rows selected.
```

Update

Existing information in the Customer table can be modified using the UPDATE statement. The WHERE condition is used to identify the particular record that needs to be changed.

```
SQL> UPDATE customer SET address = 'Hydrebad' WHERE customer_ID = 005;
1 row updated.
```